Individual Contribution Report

CMPE 321 — Project 3: Modular DBMS Engine | Boğaziçi University, Spring 2026

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Tasks & Contributions

I was responsible for Phase 4 (experimental evaluation and deliverables) and the shared document structure used across the project.

- **Phase 4 — Workload Generator.** Implemented `workload_generator.py` with four modes (`sequential`, `random`, `range`, `mixed`), a fixed `random.seed(42)` for reproducibility, and a `create type` header followed by N insert and Q query lines.
- **Phase 4 — Experiments.** Implemented `run_experiments.py` running three experiments: (1) LRU vs. MRU hit rate and I/O under sequential and random workloads; (2) `heap_scan` vs. `hash_index` vs. `bplus_tree` query cost; (3) buffer pool size sensitivity (4, 8, 16, 32, 64 frames). Collected results and wrote the analysis tables in `report.md`.
- **Deliverables.** Authored `README.md`, `record.txt` (exact reproduction commands), and the LaTeX individual contribution format used by both teammates.
- **Video.** Co-recorded the project demonstration video covering experiment methodology and live reproduction.

Collaboration & Challenges

Berkay owned Phases 0–3 while I owned Phase 4. We coordinated through `IMPLEMENTATION_GUIDE.md`, which either of us updated immediately after any design decision. The main challenge was that buffer counters accumulate across a session, so experiment scripts needed a `stats reset` between runs to isolate per-workload numbers. Understanding this required reading Berkay's counter-reset implementation, which highlighted the value of our shared working contract.

Self-Assessment

Phase 4 required experimental design thinking (controlling variables, avoiding confounds) rather than low-level data structures, which was a valuable shift in perspective. I became more comfortable writing reproducible benchmark harnesses. I would start experiments earlier in future projects to allow time to iterate on workload parameters.

Use of AI Tools

During the project I used an AI tool mainly to read up on things I was unfamiliar with. Before designing the experiments I asked it to explain what sequential flooding means in the context of buffer replacement, which helped me set up the LRU vs. MRU comparison more meaningfully. I also used it a couple of times to understand how `argparse` works and to get the LaTeX table syntax right for the report. The experiment scripts and all written content are my own work.